Lohit Gali

Automation Engineer
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Summary

- **Automation Engineer** with 5+ years of experience at the intersection of **finance, data science**, research engineering, and production-grade data systems.
- Specialized in building research pipelines, **ML-ready datasets**, and analytical tooling that support **alpha research, risk modeling, and investment decision-making**.
- Proven ability to translate quantitative research requirements into scalable, reproducible systems using **Python, SQL, cloud data platforms**, and modern ML tooling across regulated financial environments.
- Hands-on experience designing **LLM-powered agents, AI-assisted analysis workflows**, data processing, and security controls across **AWS, Azure, and enterprise environments**.
- Deep expertise in **Azure Databricks**, **PySpark, Delta Lake, and ADLS Gen2**, with a strong track record of building governed, high-reliability **ETL/ELT pipelines** for batch and streaming workloads.
- Proven ability to translate complex business and scientific workflows into optimized data models, enforce data governance with Unity Catalog, and deliver **SLA-driven pipelines through CI/CD** and monitoring-first design
- Experienced in building end-to-end applications across **frontend, backend, and cloud infrastructure**, collaborating closely with product managers, UX designers, and partner engineering teams.

Skills

AI & Agentic Systems: LLMs (OpenAI, Anthropic), Frameworks (LangChain, LlamaIndex, Hugging Face, BedRock), Prompt Engineering, Tool-Calling Agents, RAG, Vector Search
Languages & Libraries: Python, SQL, Pandas, NumPy, scikit-learn, TensorFlow, PyTorch
Machine Learning: supervised & unsupervised learning, classification, regression, clustering, model validation
Analytics & Statistics: exploratory data analysis, hypothesis testing, feature engineering, statistical modeling
Databases & Data Management: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)
Data Engineering: Databricks, Snowflake, ETL, Data Pipelines, data lakes, medallion architecture (bronze/silver/gold), ETL/ELT pipelines, Apache Kafka
Scripting & Automation: PowerShell, Python, Bash, Power Automate, VBScript, VisualCV
Data Visualization & Reporting: Power BI, Tableau, Grafana
Cloud Platforms & Services: AWS (Lambda, S3, EC2, IAM), Azure (ADLS Gen2, Fabric, Logic Apps, AD), GCP (Cloud Functions)
CI/CD & DevOps Tools: Jenkins, GitHub Actions, Azure DevOps, Docker, Kubernetes, Terraform

Professional Experience

Epik Solutions *Jul 2025 – Present*
Senior Automation Engineer (Enterprise Security & Reporting)
Project: Financial Services / Asset Reporting Modernization

- Built research-grade financial data pipelines that ingest, normalize, and version investment, operational, and control datasets to support risk analysis, trend modeling, and portfolio insights, ensuring consistency across research and reporting layers.
- Engineered ML-ready feature datasets to enable applied statistical and machine learning models for anomaly detection, forecasting, and pattern discovery across financial and operational time-series data.

- Implemented data validation, reconciliation, and monitoring frameworks to ensure numerical accuracy and auditability in regulated asset-management workflows, reducing downstream model noise and reporting discrepancies.
- Designed Python-based analytical workflows that translate complex financial questions into repeatable computations, enabling faster hypothesis testing and exploratory research.
- Developed automated KPI and performance dashboards aligned with executive and investment-risk stakeholders, transforming raw datasets into decision-ready insights.
- Applied LLM-assisted analysis pipelines to accelerate exploratory data analysis and summarization across large financial and security datasets while preserving deterministic outputs for compliance.
- Designed and automated SSRS reports integrated with Dynamics 365 (D365) to streamline operational reporting, ensuring accuracy and faster delivery of financial and business insights.

Samay Systems LLC *Jul 2024 – Jun 2025*
Senior Automation Engineer (Cloud & Compliance)
Project: Compliance Platform (SaaS) – Security & Compliance Automation

- Designed and trained supervised and unsupervised ML models addressing compliance and industry-specific analytic use cases.
- Built AI-assisted compliance workflows combining LLM reasoning with structured cloud data to guide customers through audits, investigations, and risk analysis.
- Developed Python-based automation services that collect, validate, and summarize evidence from AWS, Azure, GitHub, and Jira via secure APIs.
- Implemented RAG-style querying over historical compliance data, allowing customers to ask natural-language questions and receive contextual answers.
- Built reusable frontend and backend components to reduce duplication and improve consistency across features, accelerating delivery while maintaining quality

Arvam Tech *Jul 2023 – May 2024*
Enterprise Automation Engineer
Project: Enterprise Asset Management System

- Engineered data ingestion and transformation pipelines to consolidate asset and operational datasets into analytics-ready formats used for trend analysis and forecasting.
- Built event-driven workflows to process streaming updates and ensure near-real-time availability of analytical data.
- Delivered interactive dashboards and metrics frameworks that surfaced utilization patterns and operational risks for decision-makers.
- Developed n8n workflows to sync asset inventory and process webhook-driven events.
- Integrated Power BI dashboards with gateway refresh for asset tracking and usage metrics.
- Implemented UI enhancements and backend integrations to improve usability and reliability of internal tools

RHYM Tech *Aug 2022 – May 2023*
Associate solutions Engineer
Project: Cybersecurity Threat Intelligence Platform

- Developed analytical models using Python and SQL to analyze high-volume telemetry and event data.
- Performed data cleaning, transformation, and feature extraction to support downstream ML tasks
- Developed Python-based data pipelines to ingest high-volume event data into structured stores, enabling downstream statistical analysis and pattern detection.
- Applied time-series aggregation and anomaly detection logic to identify emerging trends across security telemetry.

- Created analytical dashboards to visualize distributions, trends, and risk indicators for technical and non-technical stakeholders.
- Supported production deployments by validating application behavior, monitoring early usage signals, and addressing issues discovered after release

ZeMoSo Technologies *Jul 2021 – Aug 2022*
Associate Developer
Project: Biotech LabOps Automation

- Developed modern web application features using HTML5, CSS, JavaScript/TypeScript, and React, ensuring responsive design and cross-browser compatibility.
- Implemented backend APIs and data access layers to support UI-driven workflows, balancing correctness, performance, and maintainability.
- Supported data science initiatives through data preparation, exploratory analysis, and basic modeling.
- Assisted in building analytical pipelines and reports aligned with business requirements.

Projects

AI-based Security Reporting Platform
Problem: Security teams relied on manual reporting and ad-hoc investigations across fragmented tools.
Solution: Built AI-driven automation and self-updating dashboards using LLM agents, APIs, and Fabric to deliver continuous insights at scale.
Agent-based Compliance Dashboard – RHYM Tech
Problem: Audit preparation and evidence validation were time-consuming and error-prone for customers.
Solution: Implemented LLM-assisted compliance workflows and RAG-based querying to automate evidence collection and analysis.
Enterprise Asset Management Automation – Arvam Tech
Problem: Manual asset tracking was inefficient.
Solution: Automated data flows with n8n, azure data factory, ADLS Gen 2, integrated Power BI, and used Power Automate for approval chains.

Education

Master's in Computer Science, Oklahoma State University, Stillwater, OK
Bachelor's in Computer Science, JNTU Hyderabad, India